

BFS of graph

Solved

Difficulty: **Easy** Accuracy: **44.09%** Submissions: **571K+** Points: **2** Average Time: **10m**

Given a **connected undirected graph** containing **V** vertices, represented by a 2-d adjacency list **adj[][]**, where each adj[i] represents the list of vertices connected to vertex i. Perform a **Breadth First Search (BFS)** traversal starting from vertex 0, visiting vertices from left to right according to the given adjacency list, and return a list containing the BFS traversal of the graph.

Note: Do traverse in the **same order** as they are in the given **adjacency list**.

Examples:

Input: adj[][] = [[2, 3, 1], [0], [0, 4], [0], [2]]

Output: [0, 2, 3, 1, 4]

Explanation: Starting from 0, the BFS traversal will follow these steps:

Visit 0 → Output: 0

Visit 2 (first neighbor of 0) → Output: 0, 2

Visit 3 (next neighbor of 0) → Output: 0, 2, 3

Visit 1 (next neighbor of 0) → Output: 0, 2, 3, 1

Visit 4 (neighbor of 2) → Final Output: 0, 2, 3, 1, 4

Input: adj[][] = [[1, 2], [0, 2], [0, 1, 3, 4], [2], [2]]

Output: [0, 1, 2, 3, 4]

Explanation: Starting from 0, the BFS traversal proceeds as follows:

Visit 0 → Output: 0

Visit 1 (the first neighbor of 0) → Output: 0, 1

Visit 2 (the next neighbor of 0) → Output: 0, 1, 2

Visit 3 (the first neighbor of 2 that hasn't been visited yet) → Output: 0, 1, 2, 3

Visit 4 (the next neighbor of 2) → Final Output: 0, 1, 2, 3, 4